

DSA Assignment

Easy

1. Reverse an Array

Write a program to reverse the elements of an array without using a built-in reverse function.

Input:

[10, 20, 30, 40, 50]

Output:

[50, 40, 30, 20, 10]

2. Count Even, Odd and Zero

Write a program to count the number of even numbers, odd numbers, and zeros in an array.

Input:

[10, 5, 0, 7, 8, 0, 13, 4]

Output:

Even: 3

Odd: 3

Zero: 2

3. Sum of Positive and Negative Numbers

Write a program to find the separate sum of positive and negative numbers in an array.

Input:

[10, -5, 20, -8, 15, -2]

Output:

Positive Sum: 45

Negative Sum: -15

Intermediate

4. Remove Duplicate Elements

Write a program to remove duplicate elements from an array.

Input:

[10, 20, 10, 30, 20, 40, 30]

Output:

[10, 20, 30, 40]

5. Find the Missing Number

An array contains numbers from 1 to N, but one number is missing. Find the missing number.

Input:

[1, 2, 3, 5, 6, 7]

Output:

Missing Number: 4

6. Rotate an Array

Write a program to rotate an array to the right by K positions.

Input:

Array: [1, 2, 3, 4, 5]

K = 2

Output:

[4, 5, 1, 2, 3]

Difficult**7. Find the Most Frequent Element**

Write a program to find the element that occurs the maximum number of times in an array.

Input:

[2, 5, 2, 8, 5, 2, 3, 5, 2]

Output:

Most Frequent Element: 2

Frequency: 4

8. Maximum Subarray Sum

Write a program to find the maximum sum of a contiguous subarray.

Input:

[-2, 1, -3, 4, -1, 2, 1, -5, 4]

Output:

Maximum Subarray Sum: 6

Subarray:

[4, -1, 2, 1]

9. Check Whether Two Arrays are Equal

Given two arrays, check whether they contain the same elements with the same frequency, regardless of their order.

Input:

Array 1: [1, 2, 2, 3, 4]

Array 2: [4, 2, 1, 2, 3]

Output:

Arrays are Equal